

Mohammad Albinhassan

PhD researcher on controlled inference, post-training, and reasoning in LLMs

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EDUCATION

Imperial College London

Department of Computing

- Advisors: Professor Alessandra Russo and Dr Pranava Madhyastha

King's College London

Department of Informatics

- **Distinction** (83% Overall)

King's College London

Department of Informatics

- **First-Class Honours** (86% Overall)

PhD Computing Research

October 2023 – Present

MSc Artificial Intelligence

September 2022 – August 2023

BSc Computer Science

September 2019 – June 2022

PUBLICATIONS

Mohammad Albinhassan, Pranava Madhyastha, and Alessandra Russo. SEM-CTRL: Semantically Controlled Decoding. *Transactions on Machine Learning Research (TMLR)*, 03/2026. [**Awarded Journal-to-Conference certification; presenting at ICML 2026, Seoul (July 2026)**].

Mohammad Albinhassan, Pranava Madhyastha, Mark Law, and Alessandra Russo. Learning and Enforcing Context-Sensitive Control for LLMs. *Association for Computational Linguistics: Student Research Workshop*.

Yaniv Aspis, **Mohammad Albinhassan**, Jorge Lobo, and Alessandra Russo. Embed2Rule - Scalable Neuro-Symbolic Learning via Latent Space Weak-Labeling. *Neural-Symbolic Learning and Reasoning: 18th International Conference, NeSy 2024*.

INDUSTRY EXPERIENCE

Research Scientist

Google DeepMind

October 2025 – January 2026

London, UK

- Contributed to frontier model research and development as a research scientist embedded with the Gemini team, including work on DeepThink, spanning model capabilities and the design of new evaluation methodologies.
- Co-led two research projects bringing together Google DeepMind researchers and Imperial faculty, PhD, and graduate students.

Research Assistant

Imperial College London

June 2024 – Present

London, UK

- Tech lead on an Imperial Policy Forum project with DSIT and the AI Security Institute (AIS), focused on LLM agents.
- Build and lead research collaborations with chemists and scientists for scientific discovery, spanning molecular and drug design, multi-step retrosynthesis, and life sciences.
- Establish departmental research collaborations, including the Thomson Reuters Lab at Imperial College London.
- Supervise internal and external research students, and monitor research projects and tool development.

Software Engineer

American Express

June 2022 – August 2022

London, UK

- Full-stack Software Engineer in the modern accounts receivable team handling credit card transactions.

Software Engineer

Intuit

June 2021 – September 2021

London, UK

- Developed integral features for the QuickBooks Payroll product, used by millions of users, in an agile scrum team.

INVITED TALKS

Fetch.ai × Google Agentic AI

University of Cambridge

March 2026

Neuro-Symbolic AI Workshop

Imperial College London and City St George's, University of London

March 2026

Scaffolds Research Group

The Alan Turing Institute

June 2025

FUNDING AND AWARDS

Academic Scholarship (Top 1%)

Imperial College London, 2023

Academic Scholarship (Top 1%)

King's College London, 2022

Undergraduate Informatics Prize

King's College London, 2022

Award for best academic performance in the Department of Informatics.

SERVICE

Conference reviewing

2023–Present

ACL, IJCAI, NeSy, and ICML.

HPC Administrator

2025–Present

Manage Imperial GPU clusters, supporting students and faculty.

RESEARCH EXPERIENCE

PhD Research

Imperial College London

- Inducing control in LLMs at both the ‘reasoning’ and ‘generation’ level, bolstering reasoning capabilities and ensuring the validity of output generations with neuro-symbolic learning (*System 2 thinking*).
- Investigating post-training mechanisms for improving reasoning in LLMs, guiding models toward more deliberate and correct reasoning trajectories.
- Conducting experiments on tasks including code generation, agent planning, semantic parsing, math, and logical problems.
- Broader interests: foundation models, reinforcement learning, neuro-symbolic learning, and reasoning in neural networks.

Transformer-Based Online Reinforcement Learning with a Shared Global Workspace

King's College London (Advisor: Dr Yali Du)

- Developed Deep Reinforcement Learning (DRL) agents using novel transformer-based architectures inspired by cognitive neuroscience (global workspace theory), outperforming state-of-the-art online DRL agents.
- Applied to discrete, continuous, lifelong, and meta-learning reinforcement learning domains and environments.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL

AI and ML: PyTorch, HuggingFace, vLLM, verl, LangChain, Weights & Biases, distributed training (multi-GPU, multi-node)

Symbolic and Reasoning: Answer Set Programming (ASP), Prolog, RDKit